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Using dispersivity values to quantify the effects of pore-scale flow focusing on enhanced reaction along a transverse mixing zone

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ABSTRACT

A key challenge for predictive modeling of transverse mixing and reaction of solutes in groundwater is to determine values of transverse dispersivity (α_T) in heterogeneous flow fields that accurately describe mixing and reaction at the pore scale. We evaluated the effects of flow focusing in high permeability zones on mixing enhancement using experimental micromodel flow cells and pore-scale lattice-Boltzmann-finite-volume model (LB-FVM) simulations. Micromodel results were directly compared to LB-FVM simulations using two different pore structures, and excellent agreement was obtained. Six different flow focusing pore structures were then systematically tested using LB-FVM, and both analytical solutions and a two-dimensional (2D) continuum-scale model were used to fit α_T values to pore-scale results. Pore-scale results indicate that the overall rate of mixing-limited reaction increased by up to 40% when flow focusing occurred, and it was greater in pore structures with longer flow focusing regions and greater porosity contrast. For each pore structure, α_T values from analytical solutions of transverse concentration profiles or total product at a given longitudinal location showed good agreement for nonreactive and reactive solutes, and values determined in flow focusing zones were always smaller than those downgradient after the flow focusing zone. Transverse dispersivity values from the 2D continuum model were between values within and downgradient from the flow focusing zone determined from analytical solutions. Also, total product and transverse concentration profiles along the entire pore structure from the 2D continuum model matched pore scale results. These results indicate that accurate quantification of pore-scale flow focusing with transverse dispersion coefficients is possible only when the entire flow and concentration fields are considered.

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1. Introduction

Accurate quantification of risk from groundwater contamination requires determining chemical concentrations downgradient from contaminant source zones. For conservative contaminants (nonsorbing and nonreacting), the only viable mechanism to reduce concentrations is by dilution of the contaminant plume with the surrounding groundwater. For (bio)reactive contaminants, degradation is controlled by the extent of mixing of substrates, and the reaction rate. For contaminants from a persistent source, degradation is thought to occur primarily along plume fringes where limiting substrates and trace nutrients mix with the contaminant plume from the surrounding groundwater transverse to the direction of flow [1–3]. In most biologically controlled degradation scenarios (e.g., in situ bioremediation), the supply of substrates to the reaction zone via transverse mixing represents the limiting time scale for degradation, and hence the reaction rate can be considered as instantaneous [4,5]. Potential improvements in understanding the mechanisms governing mixing in porous media has the potential to result in more efficient, more reliable, and lower cost remediation strategies.

Prior work indicates that transverse mixing is affected by pore geometry. Cao and Kitanidis [6] evaluated dilution of a conservative tracer in a periodic pore channel, and found that dilution (i.e., mixing) was enhanced at the necks of pores due to the focusing of streamlines, which therefore increased the mixing efficiency of molecular diffusion. In a similar study, Dykaar and Kitanidis [7] evaluated chemical degradation in a stagnant biofilm uniformly

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coating a pore surface. The authors found that degradation was enhanced at pore throats due to increased mixing of solutes. Recently, Willingham et al. [8] conducted unique microfluidic experiments and found that grain orientation also affects mixing and chemical reactions. Specifically, increased product formation rates were observed for vertically aligned ellipses compared to horizontally aligned ellipses and regularly spaced cylinders. This is primarily because flow focusing between the narrow pore throats of vertical ellipse structures enhanced mixing.

Prior work also indicates that transverse mixing is affected by permeability variations at the continuum scale. In a two-dimensional heterogeneous aquifer, Cirpka and Kitanidis [9] found that the local scale dispersion coefficient, which can be considered an accurate measure of effective mixing, increased in areas that corresponded to higher conductivity zones. The authors pointed out that it is not so much the absolute value of the conductivity as it is the contrast in conductivity with the surrounding region that controls focusing of the flow field, and hence leads to enhanced mixing. Similarly, Cirpka et al. [1] found that a conservative NaCl plume experienced enhanced dispersion in a large two-dimensional sand box experiment after passing through a high permeability lens. Permeability variations and flow focusing have also been shown to enhance rates of biodegradation in a two-dimensional flow cell experiment [10]. Werth et al. [11] evaluated the effect of flow focusing in high permeability zones on enhanced mixing and reactions in groundwater, and developed simple analytical expressions that quantify the extent of mixing and reaction enhancement. They found that mixing and reaction enhancement could be predicted based on the relative lengths and widths of high permeability inclusions in a low conductivity field, and the fraction of flow diverted through the inclusions. Based on these results, we define flow focusing as the convergence of streamlines in a flow field caused by spatially variable permeability or pore dimensions (e.g., flow focusing occurs when flow in a low permeability matrix converges in a high permeability inclusion).

In groundwater sediments, heterogeneous flow fields are common, and flow focusing and divergence occur [12]. At the pore-scale, flow focusing can result in enhanced transverse mixing and reaction of solutes along plume margins. A key challenge is determining values of transverse dispersivity in heterogeneous flow fields that accurately describe mixing and reaction at the pore scale. Historically, breakthrough concentration profiles of conservative and reacting solutes have been fit using continuum-scale models to obtain dispersivity values [13]. In many cases, transverse concentration profiles were not sufficiently resolved and transverse dispersivity values captured the effects of both mixing and spreading. Note that mixing is due to molecular diffusion, whereas spreading (i.e., hydrodynamic dispersion) is due to spatial variability in advection deforming the solute plume. Mixing contributes to reaction, whereas spreading increases the plume contact area but does not directly enhance reaction. In some cases, breakthrough profiles were highly resolved vertically (transverse to the main flow direction), such that small changes in concentration over depth were obtained [14]. In one case, transverse concentration profiles were resolved at the pore scale, and by fitting analytical solutions to transverse concentration profiles, similar dispersion coefficients were obtained from both nonreactive and reactive tracers for both homogeneous and random porous media [15]. It is not known whether this approach can be used to obtain values of transverse dispersion that accurately describe mixing and reaction along plume margins when flow focusing occurs.

The overall objective of this work is to determine if accurate values of transverse dispersivity can be obtained by continuum models fits of pore-scale results in flow focusing pore structures. We focus on comparing continuum model results with pore-scale results of mixing and reaction, where concentration profiles are resolved across individual pores. Pore-scale mixing and reaction are first quantified experimentally in two different microfluidic pore structures. These results are used to validate a lattice-Boltzmann-finite-volume model (LB-FVM), and mixing and reaction are then evaluated numerically in a larger number of flow focusing pore structures. Analytical solutions of the reactive transport equation, and a numerical solution of a two-dimensional reactive transport model that accounts for flow focusing, are then matched to concentration profiles and total product formation obtained from pore-scale simulations in order to obtain transverse dispersivity values. We demonstrate that for each pore structure, transverse dispersivity values obtained from analytical solutions of transverse concentration profiles vary along the flow path when flow focusing occurs, but that a single value of transverse dispersivity accurately captures mixing and product formation along the pore structure when the entire flow and concentration fields are considered in the two-dimensional continuum model.

2. Methods

2.1. Pore geometries

Two different pore structures were used to compare micromodel experiments with LB-FVM simulations. They each contained a preferential flow zone, but located at different longitudinal locations. The preferential flow zones were created by placing lower porosity regions on either side of a central higher porosity region. The higher porosity regions contain a uniform array of cylinders 300 μm in diameter with 40 μm pore throats. The lower porosity regions have these same cylinders and spacing, with additional smaller cylinders 120 μm in diameter added to the center of the interstitial pore space of each unit cell. For the first preferential flow case, the flow focused zone is located at the inlet (inlet focusing structure, see Fig. S1(a) in the Supplementary Materials). For the second case, the flow focused zone is downgradient from the inlet a distance of 0.465 cm (middle focusing structure, Fig. S1(b)). Preferential-flow zones are 0.92 cm long and 0.95 cm wide, and the entire pore structures are 1 cm wide by 2 cm long.

The two micromodels used in this study were fabricated at the Micro-Nano-Mechanical Systems Laboratory at the University of Illinois at Urbana-Champaign. Standard photolithography techniques described in detail elsewhere [8,16] were used to create the micromodels. High resolution masks were created by printing model designs on transparencies (5080 DPI) and transferring to soda-lime masks using photolithography. Silicon wafers were coated with a photo-resist (PR) polymer, and selectively exposed to UV light by placing each mask directly above a wafer. The areas of PR that were exposed to UV light were weakened, and were removed with a developer. Exposed areas of silicon (e.g., areas not covered by PR) were etched to a depth of approximately 30 μm using plasma generated from an ICP-DRIE (inductively coupled plasma-deep reactive ion etching). Etching of silicon with the ICP-DRIE is highly anisotropic, and creates vertical side walls with uniform flat bottoms. Flow channels were sealed by anodically bonding 0.5 mm thick Pyrex® glass to the top of each silicon wafer. Each micromodel contains a 1 cm by 2 cm porous media section.

Seven additional pore geometries (0.818 cm × 0.337 cm) were evaluated using only the LB-FVM, and the pore system dimensions are listed in Table 1. This includes one periodic array of 300 μm cylinders with 40 μm pore throats and six flow focusing cases. The periodic array represents a base case for comparison to the flow focusing cases. In the first case, the flow focusing zone is located at the beginning of the pore network, and the porosity in the low permeability zone is 0.2977. In the second case, the flow focusing zone is equivalent in size, but has been moved downgra-

Table 1
Pore-structure parameters for the LB-FVM simulations.

Description	Distance (cm)	Width mix (cm)	Length mix (cm)	Interstitial diameter (μm)	Darcy velocity (cm/s)	Porosity (–)
Homogeneous	NA	NA	NA	NA	0.01134	0.3913
Inlet	0.025	0.098	0.193	120	0.01121	0.372
Middle short	0.170	0.098	0.143	120	0.01132	0.3763
Middle medium	0.170	0.098	0.193	120	0.01133	0.372
Middle long	0.170	0.098	0.240	120	0.01137	0.3676
Middle narrow	0.170	0.048	0.240	120	0.01134	0.3629
Middle focus	0.170	0.098	0.240	135	0.01153	0.3583

Distance is length downgradient from the inlet where the preferential mixing zone starts. Width mix and length mix refer to mixing zone width and length, respectively. Interstitial diameter is the diameter of the smaller interstitial grains used to create the preferential focused zone.

dient. In the third and forth case, a shorter and longer mixing zone were evaluated, respectively. In the fifth case, the focusing zone width was decreased. In the sixth and final case, flow focusing was evaluated by using larger interstitial grains to reduce porosity of the low permeability zone to 0.2626. In all six flow focusing pore structures, the porosity of high permeability regions are the same as the base homogeneous case. The geometries of all six flow focusing pore structures are illustrated in Fig. 1.

2.2. Pore-scale experiments

In all experiments, the fluorophore Oregon Green Bapta 488 (OG5N) in the Hexapotassium salt form was used (Invitrogen Inc.). OG5N reacts with Ca^{2+} (hereafter referred to as Ca) in a 1:1 stoichiometric ratio to form a fluorescent product. OG5N in the unbound form (e.g., before binding with Ca) has very low background intensity that increases approximately eight fold when bound with Ca. To evaluate reactive transport in the two micromodels, Ca and OG5N were introduced via two separate and parallel fluid streams. The chemicals mixed within the porous media due to molecular diffusion and reacted. The reaction rate for OG5N and Ca is fast compared to advection and diffusion time scales [8]. As a result, product formation in the micromodels is directly related to mixing of the substrates and can be evaluated independently of reaction rate. To conduct conservative tracer experiments in the two micromodels, OG5N was premixed with Ca and the resulting fluorescent product solution was subsequently injected through one inlet port as the conservative tracer while MillQ water was injected through the other. Two ISCO syringe pumps were used to inject all solutes; these pumps are driven by direct current and are pulseless. Molecular diffusion coefficients in water at infinite dilution for Ca, OG5N, OG5N + Ca are 7.9×10^{-6} cm^2/s, 3.6×10^{-6} cm^2/s, and 3.5×10^{-6} cm^2/s, respectively [8].

To determine chemical concentrations within micromodels, fluorescent microscopy was used to quantify fluorescent intensity, which is related to concentration through a calibration curve. A Nikon Epiphot 200 microscope with a Prior motorized stage was used to capture steady-state fluorescent images, that were montaged together to form a single image which captures the entire mixing zone region. A total of approximately 30 images was captured to form each single montaged image. Further details of the experimental approach and calibration method are reported in Willingham et al. [8].

2.3. Pore-scale modeling

To model pore-scale reactive transport we use the lattice-Boltzmann method to solve for interstitial pore velocities, and a finite-volume model to solve for reactive transport. An advantage of using the LB-FVM approach is that the only input parameters are the volumetric flow rate, pore structure, molecular diffusion coefficients, and intrinsic reaction rate. The LB method is a kinetic based theory for modeling momentum and solute transport [17] and is naturally suited to handle arbitrarily complex boundary conditions, thereby making it attractive for modeling pore-scale transport problems. The LB fluid flow evolution equation has been shown to recover the continuity and Navier–Stokes equations [18].

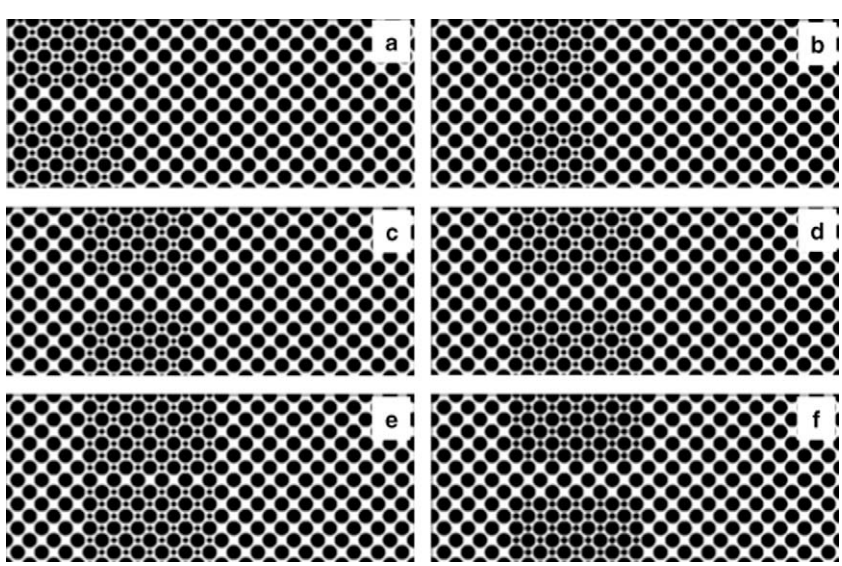

Fig. 1. Flow focusing pore structures: (a) Inlet, (b) middle short, (c) middle medium, (d) middle long, (e) middle narrow and (f) middle focus.

At the inlet and outlet, fluid pressure is specified such that the resulting Darcy velocity is either equivalent to the velocity in the micromodel experiments, or is set to a constant Darcy flow rate of 0.0111 cm/s. No slip boundary conditions are imposed at all solid grains via a bounce-back boundary condition. The node spacing is set to 5 μm for all LB simulations.

The general equations for describing steady-state transport of solutes A and B at the pore scale, that undergo second order reaction $A + B \rightarrow P$, can be written as

$$0 = D_A \nabla^2 A - \nabla \bullet (\vec{u}A) - kAB \tag{1}$$

$$0 = D_B \nabla^2 B - \nabla \bullet (\vec{u}B) - kAB \tag{2}$$

$$0 = D_P \nabla^2 P - \nabla \bullet (\vec{u}P) + kAB \tag{3}$$

where k is the intrinsic reaction rate, D_A, D_B, and D_P are the molecular diffusion coefficients of reactants A and B and product P, respectively, and $\vec{u}$ is the steady-state velocity field obtained from the LB simulations.

The finite volume method was used to discretize equations (1)–(3) using the same grid spacing as for the LB simulations (5 μm). For reactive experiments, the inlet boundary condition (BC) was defined as $B_{\text{in}} = B_o$ and $A_{\text{in}} = 0$ for the top half of the inlet flow boundary and as $A_{\text{in}} = A_o$ and $B_{\text{in}} = 0$ for the bottom half. The outlet BC was defined as pure advection with the gradient in the x direction equal to zero. In all cases, k was set to a large value to approximate an instantaneous reaction, and values of D_A, D_B, and D_P were those for OG5N, Ca, and OG5N + Ca. Since D_A and D_P only differed by 1.4%, the same value was used to facilitate evaluation of pore-scale results with analytical solutions. For nonreactive experiments (see Fig. S1), the inlet boundary conditions were defined as $P_{\text{in}} = P_0$ for the top half and $P_{\text{in}} = 0$ for the bottom half. Along all the internal boundaries with the solid grains, zero flux was imposed.

The equations were solved using a Newton Raphson approach, in which the reaction term was linearized and solved iteratively. Pore-scale modeling results are compared to micromodel results by converting model concentration values into intensity values using the calibration profile. Intensity values from micromodel experimental results could not accurately be converted to concentration values because the background intensity signal from OG5N overlaps with the intensity signal from OG5N + Ca in the pore structure. Further details of the pore-scale modeling approach are given in Willingham et al. [8].

2.4. Determining transverse dispersivity values

2.4.1. Analytical solutions

Nonreactive solute concentration profiles transverse to the flow direction from LB-FVM simulations were fit to an analytical solution of the advection–dispersion equation to obtain transverse dispersion coefficients. We assume a continuum with a uniform velocity; this assumption is made because in the field, details of the flow field are typically not available. The corresponding equation is the well-known error function solution [19]

$$P(x,y) = \frac{1}{2} P_0 \left[erfc\left(\frac{y}{2\sqrt{D_{T,NR} x/v}} \right) \right] \tag{4}$$

where y is the transverse direction with a value of zero at the mixing centerline, P_0 is the concentration of the nonreactive tracer input along the top half of the inlet boundary, $D_{T,NR}$ is the nonreactive transverse dispersion coefficient, x is the distance from the inlet and v is the average linear velocity. (Note that the more familiar form of Eq. (4) is for transient diffusion where x/v represents time.) For each pore structure, a transverse concentration profile in the flow focusing zone, and a transverse concentration profile downgradient from the flow focusing zone (i.e., after streamlines diverged), were considered. Also, point-wise (i.e., grid-scale) concentration profiles were used to avoid any artificial smearing. Point-wise profiles have data gaps where grains are present, so only data between grains was used for fitting. This approach was shown to yield accurate nonreactive transverse dispersion coefficients for uniform, random, and anisotropic pore structures [15]. Acharya et al. [15] also demonstrated that values of $D_{T,NR}$ found by fitting Eq. (4) to point-wise concentration profiles are similar to values found from spatial moment analysis. Values of $D_{T,NR}$ were converted to transverse dispersivities using the following equation

$$D_{T,NR} = \alpha_{T,NR} v + D_{mol} \tau \tag{5}$$

where $\alpha_{T,NR}$ is the transverse dispersivity for nonreactive experiments, and τ is the estimated tortuosity factor set equal to porosity [20].

Reactive solute concentration profiles transverse to the flow direction from LB-FVM simulations were fit to analytical solutions of the advection–dispersion–reaction equation to obtain transverse dispersion coefficients. In order to obtain the analytical solution, we assume a continuum with a uniform velocity and an instantaneous reaction. In the case of an instantaneous reaction, all of the reaction is concentrated at the location where the reactants A and B mix due to molecular diffusion. If we further assume that the diffusion coefficient of the product is equal to that of one of the reactants (in this case, we assume $D_P = D_A$), then the analytical solution for the product is given as:

$$P(x,y) = \frac{A_0}{2} erfc\left(\frac{y}{2\sqrt{D_{T,R} x/v}} \right) - \frac{A_0}{1 + erf\left(\frac{y_*}{2\sqrt{D_{T,R} x/v}} \right)} \left[erf\left(\frac{y_*}{2\sqrt{D_{T,R} x/v}} \right) - erf\left(\frac{y}{2\sqrt{D_{T,R} x/v}} \right) \right] \quad \text{for } y < y^* \tag{6}$$

$$P(x,y) = \frac{A_0}{2} \left[erfc\left(\frac{y}{2\sqrt{D_{T,R} x/v}} \right) \right] \quad \text{for } y > y^* \tag{7}$$

where y^* represents the location of the reaction zone, and all other parameters are defined above. The reaction zone location is given by

$$y^* = \alpha \sqrt{x/v} \tag{8}$$

where the parameter α is determined from solving an implicit equation that depends upon the ratio of the diffusion coefficients, D_A/D_B, and the ratio of the inlet concentrations, $A_{\text{in}}/B_{\text{in}}$ [21,22]. The solution (6) is reported by Nambi et al. [21] and was derived by modifying the solution presented earlier in [22] to account for modified boundary conditions. For the simple case of equal diffusion coefficients and equal inlet concentrations, $\alpha = 0$ (i.e., the reaction zone is along the centerline as would be expected). As with the nonreactive case, transverse concentration profiles in both flow focusing zones, and downgradient from flow focusing zones, were considered, and point-wise (i.e., grid-scale) concentrations profiles were used to avoid artificial smearing.

Reactive solute concentration profiles transverse to the flow direction from LB-FVM simulations were also integrated along the transverse direction at each grid point along the direction of flow to obtain total product formed as a function of longitudinal distance. The corresponding analytical solution was used to fit the LB-FVM results, and is given as

$$m(x) = 2A_0\theta \sqrt{\frac{D_{T,R,P} x/v}{\pi}} \exp\left(-\frac{\alpha^2}{4D_{T,R,P}} \right) \frac{1}{1 + erf\left(\frac{\alpha}{2\sqrt{D_{T,R,P}}} \right)} \tag{9}$$

where $m(x) = \int P(x,y)dy$, θ is the porosity, and all other parameters were defined above. The total product formed in both flow focusing zones, and downgradient, were fit separately to obtain two different transverse dispersion coefficients. This same approach was also used to obtain accurate reactive transverse dispersion coefficients for random pore structures [15]. All reactive transverse dispersion coefficients were converted to transverse dispersivity values using Eq. (5), with $D_{T,R}$ and $\alpha_{T,R}$ used when transverse concentration profiles are fit (Eqs. (6)–(8)), and $D_{T,R,P}$ and $\alpha_{T,R,P}$ used when the total product at any longitudinal location is fit (Eq. (9)).

2.4.2. 2D Numerical continuum model simulation

Transverse dispersion coefficients for reactive cases were also determined with a two-dimensional continuum-scale model that simulates flow and reaction in porous media with flow focusing. Two hydraulic conductivity zones for the continuum model were created to mirror the size and porosity distributions in each of the pore-scale systems; i.e., two low permeability regions at the inlet or middle of the porous media focused flow in higher permeability media. The hydraulic conductivity ratio between the low and high permeability regions was determined by computing the velocity field across uniform pore structures of the high and low porosity structures via the LB method under the same pressure gradient, and the absolute value of the high permeability structure was estimated using the Kozeny–Carman correlation [20].

Flow in the continuum model system was solved using the two-dimensional equation of continuity. Reactive transport was solved using the same equations (Eqs. (1)–(3)) and the finite volume technique used for pore-scale modeling, except that dispersion coefficients were substituted for diffusion coefficients, and they were defined as

$$D_{L,R,C,i} = \alpha_{L,R,C}\,v + D_{mol,i}\tau \tag{10}$$

$$D_{T,R,C,i} = \alpha_{T,R,C}\,v + D_{mol,i}\tau \tag{11}$$

where i represents either reactant A, reactant B, or product P, $D_{L,R,C}$ and $D_{T,R,C}$ are the longitudinal and transverse dispersion coefficients fitted with the 2D reactive continuum model, $\alpha_{L,R,C}$ and $\alpha_{T,R,C}$ are the corresponding longitudinal and transverse dispersivities, and other parameters were previously defined. The grid spacing is equal to 10 μm; this is twice as big as the pore-scale LB-FVM model. Grid spacings of 5 μm, 10 μm and 20 μm were compared for the inlet focusing pore structure with no effect on the results. Longitudinal dispersivity (α_L) equal to grain diameter was used [23–26] and the transverse dispersivity ($\alpha_{T,R,C}$) was adjusted for each pore structure to minimize the sum of the squared errors between the total product formed as a function of longitudinal distance calculated from the two-dimensional continuum-scale model, and from the LB-FVM. Values of $\alpha_{L,R,C}$ and $\alpha_{T,R,C}$ in the two low conductivity blocks were determined based on the hydraulic conductivity ratio [2,26–28], so the only fitting parameter is $\alpha_{T,R,C}$. In contrast to model fits using the analytical solutions, the fit was evaluated over the entire length of the pore structure (both inside and outside of the flow focusing region).

3. Results

3.1. Direct comparison of micromodel experiments and LB-FVM simulations

The OG5N + Ca concentration is linearly correlated to signal intensity. Therefore, a calibration curve was obtained by injecting known concentrations of OG5N + Ca through the micromodel and measuring corresponding fluorescence intensities; the corresponding linear relationship was used for converting concentrations of OG5N + Ca from LB-FVM simulation results to signal intensities [8]. A direct comparison of intensity distributions from micromodel and LB-FVM results with the inlet focusing structure is shown in Fig. 2 for the reactive solutes. The two initially vertically segregated solutes mix within the porous media due to transverse diffusion, and react to form a product that is seen as the higher intensity region along the centerline. Results are compared at greater resolution by plotting intensity values from experimental and model results at two different cross sections in Fig. 3. There is good agreement between modeling and experiments (the asymmetric concentration/intensity profile is primarily attributed to higher molecular diffusivity of Ca, which caused more product to form along the OG5N side; this asymmetry is also predicted by the analytical solution given in Eqs. (6)–(8)). Similar comparisons were made for the nonreactive solute in the same pore structure (Figs. S2 and S3), and for the reactive and nonreactive solutes in the middle focusing structure (results not shown). In all cases, agreement between model and experimental intensity distributions and cross sectional profiles was good.

We also compared the total intensity (i.e., intensity integrated transverse to flow at each grid point and averaged over 1.5× grain diameters in the longitudinal direction to smooth the profile) versus longitudinal distance in both experimental pore structures

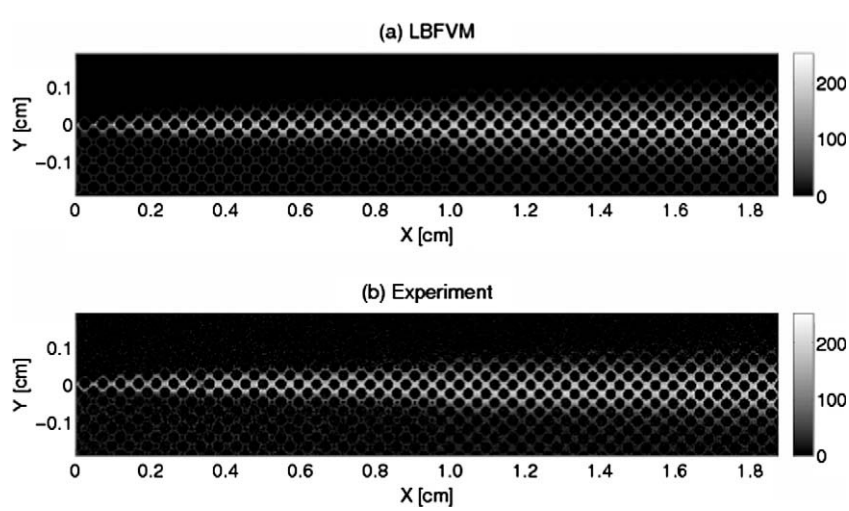

Fig. 2. Direct comparison of (a) intensity measurements from an experimental micromodel flow cell and (b) calibrated intensity values from the LB-FVM simulation, both in the inlet pore structure (Fig. 1a)

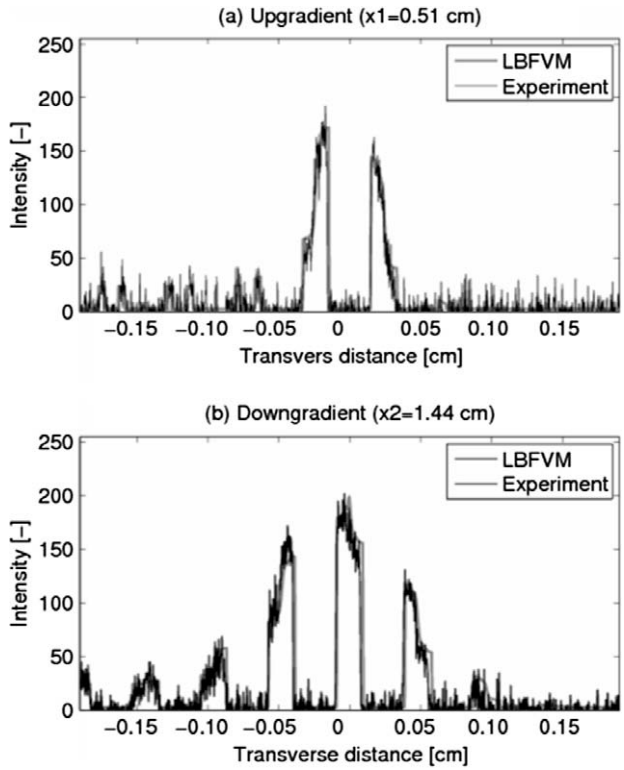

Fig. 3. Cross sections comparing intensity values from a micromodel experiment and a LB-FVM simulation, both in the inlet pore structure. Cross sections are taken from images shown in Fig. 2.

with LB-FVM results; the results are shown in Fig. 4. The integrated intensity value at the inlet is above zero due to the background fluorescent intensity of OG5N, and the value is lower for the inlet pore structure due to lower porosity near the inlet. In both cases there is good agreement (within 10%), and the intensities increase with distance due to the formation and transport of product along the mixing line. Agreement among all experimental and model comparisons indicates that the LB-FVM adequately captures the mechanisms that control pore-scale flow focusing under both non-reactive and reactive conditions. These results support the findings from our earlier work [8], where we found that the LB-FVM accurately captures the governing physics in micromodel experiments with a variety of pore geometries containing regularly spaced cylinders and ellipses.

3.2. Effect of flow focusing pore structure on enhanced mixing and product formation

To evaluate the effect of flow focusing regions on transverse mixing and reaction, the LB-FVM was used to evaluate product formation for the base case (uniform array of cylinders), and the six preferential mixing pore structures (i.e., Fig. 1). Results of product concentration versus distance and mass flux versus distance are compared in Fig. 5. Concentration values represent an average at each cross section location (i.e., average concentration over transverse cross section 1.5 grain diameters wide). Mass flux values were determined by multiplying local concentration values by local velocities (diffusive mass flux is neglected due to the high flow velocities, i.e., $Pe > 270$, considered in this study), and then summing mass flux values at each cross section location. The rate of increase in concentration values drops in regions of flow focusing, and in some cases becomes negative. In contrast, mass flux values increase more in regions of flow focusing than outside. Also, final mass flux values for flow focusing pore structures are always greater than for the base case. Concentration values drop in regions of flow focusing because concentrations in individual streamlines do not appreciably change, but the relative area (in 2D) of streamlines containing product that are focused decreases relative to streamlines without product that are not focused at any particular cross sectional location. However, mass flux values show that more mixing and product formation occur inside flow focusing regions, and the extent of mixing and reaction exceed that when flow focusing does not occur.

Comparison of mass flux values shows that total product formation is similar for the inlet and middle medium cases (see Table 1 for pore dimensions). These two cases have the same mixing zone widths and lengths, indicating that the longitudinal location of the flow focusing zone plays little to no role in the extent of mixing and product formation. Comparison of mass flux values also shows that total product formation increases from the middle short, to the middle medium, to the middle long (it should be noted that the total volumetric flowrate is the same for all the different cases considered in Fig. 5.). The mixing zone lengths increase in the same order, indicating that with greater length of flow focusing zones,

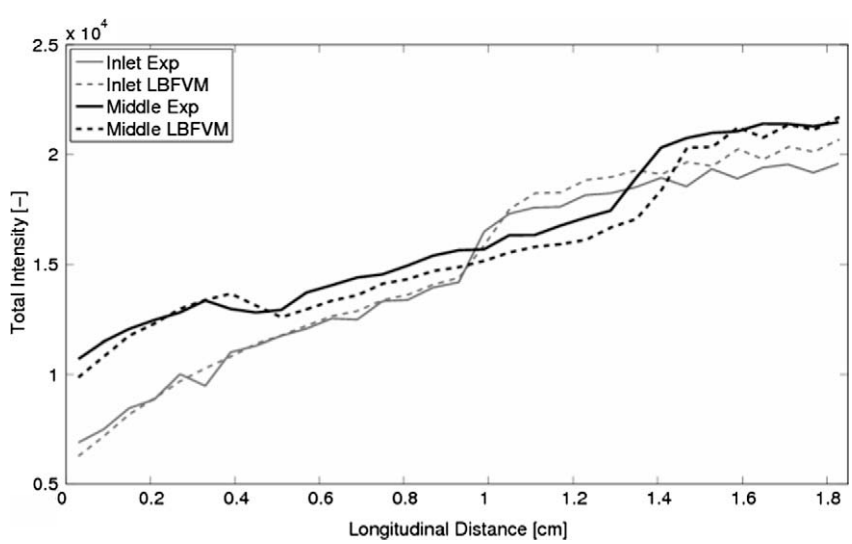

Fig. 4. Comparison of intensity values versus distance from micromodels and LB-FVM simulations in both inlet and middle focus pore structures.

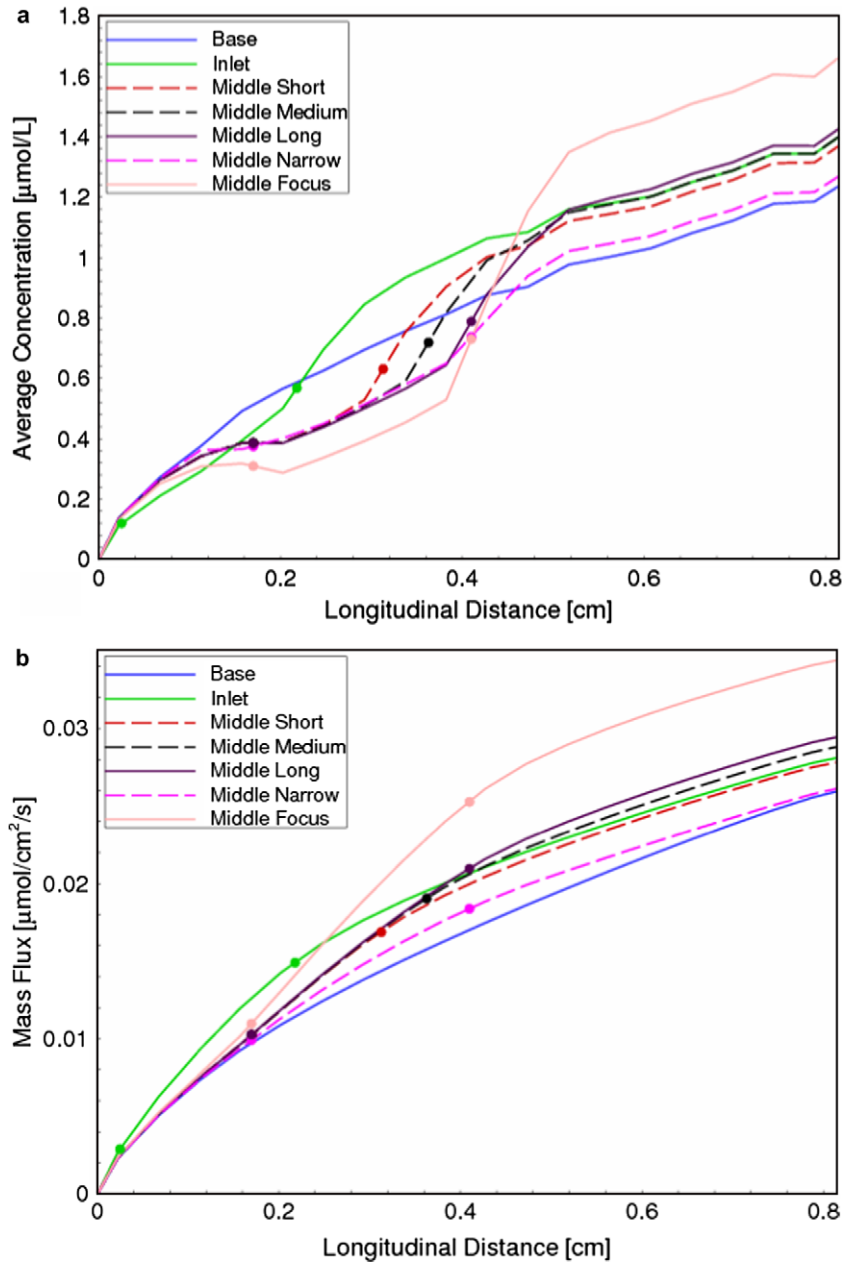

Fig. 5. (a) Average concentration and (b) mass flux versus distance plots for all seven LB-FVM pore structures (start and end of the focusing zones noted by dots). Base represents the uniform array of cylinders.

the extent of mixing and product formation increases. Comparison of final mass flux values (i.e., value at the end of the pore structure) for the middle focused and middle long cases indicates greater product formation in the former; the porosity contrast is greater in the former resulting in greater flow focusing (Fig. S4). Hence, greater flow focusing enhances mixing and product formation. This confirms the continuum results by Cirpka and Kitanidis [9] and Werth et al. [11], which indicate that product enhancement is proportional to the relative contrast between the permeability of the higher conductivity lens and the lower permeability surrounding material. Last, the final mass flux value for the middle narrow is the lowest among the preferential flow structures considered, and similar to the base case. Detailed evaluation of the velocity field for this case (Fig. S4) indicates that there is relatively little flow focusing compared to the other structures; this is because the flow focusing region is too narrow (i.e., only one grain diameter wide) to induce streamline compression.

3.3. Effect of pore-scale flow focusing on transverse dispersivity values

3.3.1. Analytical solution results

Analytical solution fits of both nonreactive and reactive solute profiles from pore-scale model results for the inlet pore structure case are shown in Fig. 6. Cross section concentration profiles for the nonreactive tracer in the flow focusing zone, and downgradient from the flow focusing zone, are shown in Fig. 6a.

Data gaps are apparent due to the presence of grains. Model fits coincide with the data, and fitted transverse dispersivity values are listed in Table 2. Model fits for other pore structure cases are not shown, but corresponding values of $\alpha_{T,NR}$ are listed in Table 2. The value of $\alpha_{T,NR}$ for the base case is equal to the value obtained by Acharya et al. [15] for the same pore structure. Values of $\alpha_{T,NR}$ in the flow focusing zone are 1.9–6.2 times smaller than downgradient from the flow focusing zone, indicating the influence of streamline compression and expansion.

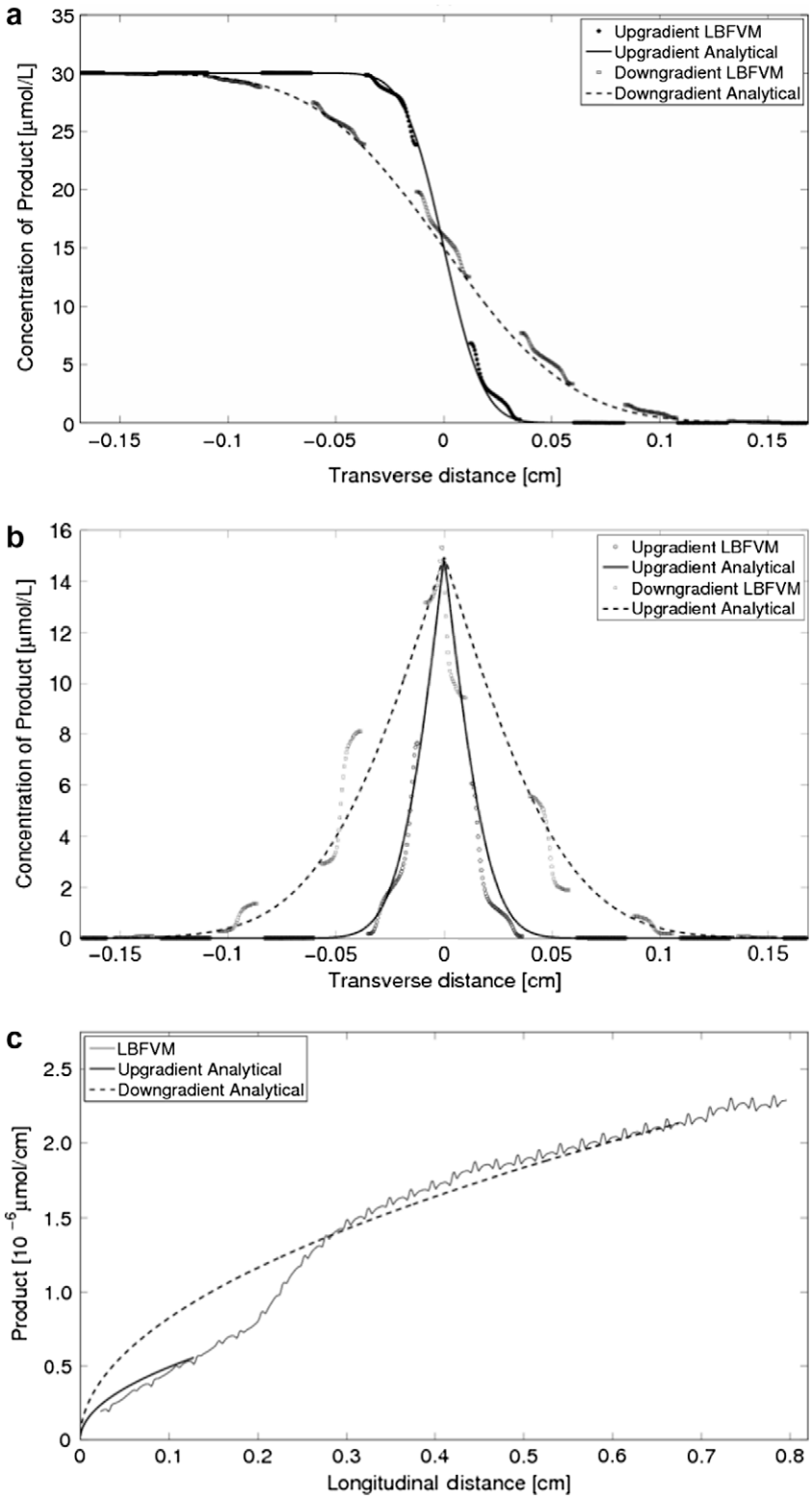

Fig. 6. Analytical solution fits of (a) transverse concentration profiles for the nonreactive tracer both in and downgradient from the flow focusing zone, (b) transverse concentration profiles for the reactive tracer both in and downgradient from the flow focusing zone, and (c) total product as a function of distance, all from the pore-scale model results for the inlet pore structure.

Reactive tracer results are shown for both the upgradient and downgradient transverse concentration profiles (Fig. 6b), and for total product formation (i.e., $m(x)$) (Fig. 6c); as before, separate fits are obtained in the flow focusing zone and downgradient. For transverse concentration profiles, analytical solution results capture the overall data trends well. However, discrepancies are apparent due to irregular concentration profile shapes between grains. For total product formation, analytical solution results coincide with pore-scale data in the regions of interest (i.e., either in the flow focusing zone or downgradient). It is clear that one analyt-

Table 2
List of pore structures and associated values ($\times 10^{-4}$ cm) of $\alpha_{T,NR}$ (transverse dispersivity of nonreactive solutes), $\alpha_{T,R}$ (transverse dispersivity of reactive solutes) and $\alpha_{T,R,P}$ (transverse dispersivity of product formed from reactive solutes) determined from analytical solutions, and from continuum model fits ($\alpha_{T,R,C}$) of pore-scale model results.

Description	$\alpha_{T,NR}$		$\alpha_{T,R}$		$\alpha_{T,R,P}$		$\alpha_{T,R,C}$
	$x1$	$x2$	$x1$	$x2$	$x1$	$x2$	
Base	11.25	12.04	11.47	11.41	10.25	11.79	10.40
Inlet	8.44	15.76	8.07	15.40	6.09	17.36	9.86
Middle short	6.22	15.15	5.95	14.14	5.23	16.14	10.38
Middle medium	6.10	15.73	5.81	14.64	5.32	17.38	10.60
Middle long	6.02	15.97	5.73	15.48	5.40	18.57	10.76
Middle narrow	5.74	12.11	5.67	11.57	5.94	14.72	12.30
Middle focus	3.68	22.96	3.36	21.21	3.10	27.18	9.70

$x1$ = 0.2255 for all cases except inlet ($x1$ = 0.1285); $x2$ = 0.6745.

ical solution profile will not match the entire total product profile obtained from pore-scale flow focus model results.

Values of transverse dispersivity for the reactive solutes are listed in Table 2, along with values for the remaining pore structure cases. As before, values of $\alpha_{T,R}$ in flow focusing zones are less than values downgradient from the flow focusing zone (for all flow focusing pore structures, $\alpha_{T,R}$ values in the flow focusing zone are 1.9–6.3 times lower than downgradient), indicating the influence of streamline compression and expansion. Also, compared to the middle long pore structure, lower porosity in the middle focus pore structure resulted in more streamline compression in the focusing zone, and more expansion downgradient from the focusing zone (Fig. S4 in the Supplemental Materials). This resulted in a lower dispersivity value in the focusing zone and higher dispersivity value downgradient. Values obtained from reactive transverse concentration profiles ($\alpha_{T,R}$) are similar to those obtained from the nonreactive transverse concentration profiles ($\alpha_{T,NR}$); in the flow focusing zone and downgradient the corresponding values are all within 9%. Values obtained from total product formation ($\alpha_{T,R,P}$) are also similar to those obtained from reactive transverse concentration profiles ($\alpha_{T,R}$); the biggest difference is 28% and most values (all except three) are within 20%.

3.3.2. 2D numerical model simulation results

Comparison of longitudinal and transverse velocities for the inlet pore structure from the pore- and 2D continuum-scale models showed good agreement, indicating that flow at the pore scale can be adequately represented by the 2D continuum-scale model. Contour profiles of concentration from both the pore- and 2D continuum-scale numerical models are shown in Fig. 7 for the inlet pore structure case. The contour profiles of concentration show qualitative agreement; comparison of transverse concentration profiles from pore- and 2D continuum-scale models also showed good agreement and the results (data not shown) are similar to Fig. 6b.

To better compare pore- and 2D continuum-scale model results, total product formation (i.e., integrated over the transverse direction) versus longitudinal distance is compared for the optimal transverse dispersivity in Fig. 8, again for the inlet pore structure case. The profiles show excellent agreement over the entire system length. Agreement was similar for other pore structure cases, and the corresponding values of $\alpha_{T,R,C}$ are listed in Table 2. All $\alpha_{T,R,C}$ values except the middle narrow case (18% higher) are within 7% of the base case. The value of $\alpha_{T,R,C}$ for the base case is within 9% of the average value of the corresponding base case values for

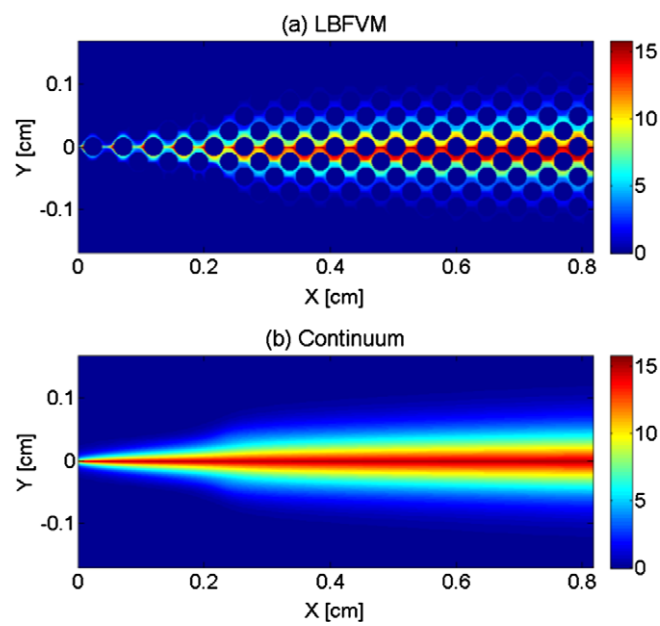

Fig. 7. Comparison of contour profiles of concentration from (a) pore- and (b) 2D continuum-scale numerical models, both for the inlet pore structure case.

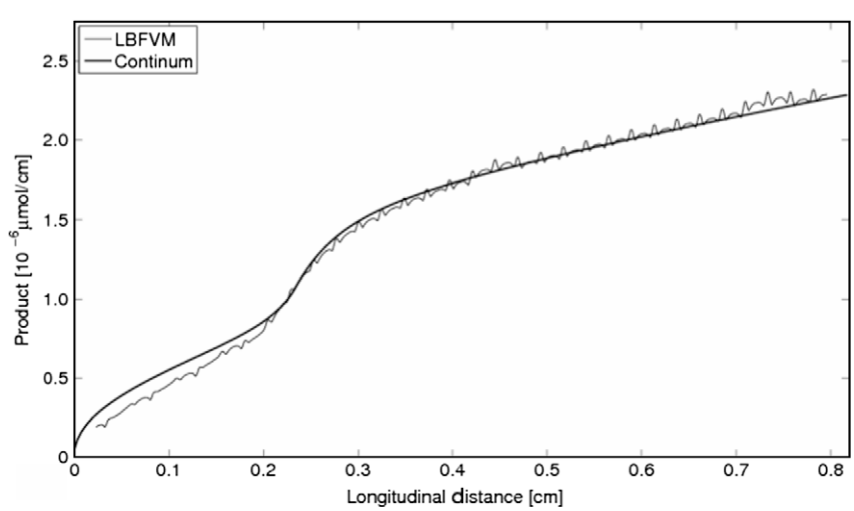

Fig. 8. Comparison of total product formation along the direction of flow from pore- and 2D continuum-scale models, both for the inlet pore structure case.

$\alpha_{T,NR}$, $\alpha_{T,R}$ and $\alpha_{T,R,P}$. Values of $\alpha_{T,R,C}$ in all flow focusing pore structures fall between values of $\alpha_{T,NR}$, $\alpha_{T,R}$ and $\alpha_{T,R,P}$ determined in the flow focusing zone and downgradient. This indicates that values of α_T determined from cross section concentration profiles based on the assumption of uniform velocity are not accurate due to the convergence and divergence of streamlines from flow focusing, and that knowledge of both the flow and concentration field is required for accurate determination of α_T.

4. Discussion

We evaluated the effects of preferential flow focusing on mixing enhancement using pore-scale LB-FVM simulations and experimental micromodel flow cells. We directly compared micromodel results with LB-FVM simulations using two different flow focused pore structures. We obtained excellent agreement between the two indicating that the LB-FVM captures the pore-scale behavior observed in the flow focused micromodel experiments.

Six different flow focusing pore structures were then systematically tested using LB-FVM to determine the effects of flow focusing location, length, width, and porosity contrast on product formation. The average linear velocities for the different flow focusing pore structures are 5–12% higher than the base case. High velocities result in shorter residence times and less contact time between the two reactants, which would result in less product; however, simulation results indicate that the extent of reaction for flow focused scenarios increased by up to 40% compared to identical structures not containing the flow focused region. Simulations evaluating the effect of flow focusing location on extent of product formation demonstrated that the longitudinal location does not have a significant effect on the overall extent of reaction. The length of the flow focusing region has a measurable but small effect on the overall extent of product formation. In agreement with previously published continuum studies [9,11], our simulations demonstrate that the larger the relative amount of fluid focused into the preferential flow zone (e.g., the larger the difference in permeability between the inclusion and matrix) the larger mixing enhancement.

Analytical solutions for nonreactive tracer spreading, and for product formation from reactive solutes were used to obtain transverse dispersivity values from pore-scale concentration data. Transverse dispersivity values determined from transverse concentration profiles of a nonreactive tracer, and from transverse concentration profiles of product from reacting solutes, showed good agreement. Similarly, Acharya et al. [15] found that values of transverse dispersion obtained from nonreactive and reactive solutes were identical. Also, transverse dispersivity values obtained by fitting data from flow focusing zones are always smaller than those determined by fitting data from sections downgradient of the flow focusing zone. A 2D continuum model with hydraulic conductivity values assigned to match low and high porosity zones was also used to simulate the pore-scale results, and one value of the transverse dispersivity accurately captured total production formation along the entire pore structure. This indicates streamline convergence in flow focusing zones compresses transverse concentration measurements, and results in artificially low transverse dispersivity values (if a uniform flow field is considered), and streamline divergence after flow focusing zones expands transverse concentration measurements, and results in artificially high transverse dispersivity values.

These findings show that pore-scale flow focusing is an important mechanism for mixing diffusion limited substrates under laminar flow conditions, and the accurate quantification of pore-scale flow focusing with transverse dispersivity values is possible for the conditions studied, provided the entire flow field and concentration profiles are known. The findings also suggest that transverse dispersivity cannot be accurately quantified by fitting vertical measurements of concentration to model fits when heterogeneity is present that results in flow focusing.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.advwatres.2010.02.004.

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